

Stat 486

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HomeWork-1

Problem 1 Let  $V$  be the number of false rejections &  $R$  be the total number of rejections

$$\frac{V}{R} = 0 \text{ where } R = 0$$

The FDR is

$$FDR = E \left[ \frac{V}{R} \right]$$

For  $R \geq 1$ , we are given that

$$\frac{V}{R} \leq 1 \text{ (} V \geq 1 \text{)}.$$

So, when  $R = 0$ , then  $V = 0$  so  $V/R \neq 0$

$V/R = 0 \leq 1$  ( $V \geq 1$ ) also holds.

So Overall,

$$\frac{V}{R} \leq 1 \text{ (} V \geq 1 \text{)} \Rightarrow E \left[ \frac{V}{R} \right] \leq E [1 \text{ (} V \geq 1 \text{)}]$$

That is

$$FDR \leq FWER.$$

Therefore if  $FWER \leq \alpha$ , then  $FDR \leq \alpha$ .

Problem 1/b) Let  $A_j$  be the event "we falsely reject hypothesis  $j$ ". (i.e. Reject  $H_{0j}$  when it is actually true). Then.

FWER is the probability of at least one false rejection  $\rightarrow$

$$\text{FWER} = P\left(\bigcup_{j=1}^M A_j\right)$$

By the union bound (Bonferroni inequality),

$$P\left(\bigcup_{j=1}^M A_j\right) \leq \sum_{j=1}^M P(A_j)$$

If the test for hypothesis  $j$  controls Type I error at level  $\alpha_j$ , then  $P(A_j) \leq \alpha_j$ . So

$$\text{FWER} \leq \sum_{j=1}^M \alpha_j$$

Problem 1/c) Test each at level  $\alpha = 0.05$

Reject all with  $p\text{-value} \leq 0.05$

• Reject:  $H_{01}, H_{02}, H_{03}, H_{08}, H_{09}, H_{10}$

• Do not reject:  $H_{04}, H_{05}, H_{06}, H_{07}$

ii) Control FWER at  $\alpha = 0.05$  (choose Bonferroni)

Bonferroni uses threshold  $\alpha/M = 0.05/10$   
 $= 0.005$

Reject  $p$ -values  $\leq 0.005$

$\cdot H_{10} (0.0009), H_{01} (0.0011), H_{09} (0.004)$

So reject:  $H_{01}, H_{09}, H_{10}$ .

iii) Control FDR at  $q = 0.05$

Sort  $P$  Values:  $\rightarrow$

- 1) 0.0009 ( $H_{10}$ )
- 2) 0.0011 ( $H_{01}$ )
- 3) 0.004 ( $H_{09}$ )
- 4) 0.006 ( $H_{08}$ )
- 5) 0.017 ( $H_{03}$ )
- 6) 0.031 ( $H_{02}$ )
- 7) 0.07 ( $H_{07}$ )
- 8) 0.11 ( $H_{05}$ )
- 9) 0.32 ( $H_{04}$ )
- 10) 0.90 ( $H_{06}$ )

BH threshold are  $(k/M)q = (k/10) \cdot 0.05$

They are 0.005, 0.01, 0.015, 0.02, 0.025, ...

Largest  $k$  with  $p(k) \leq (k/10)0.05$  is  $k=5$   
because  $0.017 \leq 0.025$  but  $0.031 \not\leq 0.03$

So reject :  $H_{10}, H_{01}, H_{09}, H_{08}, H_{03}$ .

iv) Control FDR at  $q = 0.2$  (BH)

Threshold  $(k/10) \cdot 0.2$  : 0.02, 0.04, 0.06, 0.08,  
0.10, 0.12, 0.14, 0.16, 0.18,  
0.20.

Largest  $k$  is  $k = 8$  because  $p(8) = 0.11 \leq 0.16$   
but  $p(9) = 0.32 \not\leq 0.18$ .

So reject :  $H_{10}, H_{01}, H_{09}, H_{08}, H_{03}, H_{02}, H_{07}, H_{05}$ .

v) Among the rejections at  $q = 0.2$ ,  
approximately how many false positives?

FDR is  $E[V/R]$

If BH Control s FDR at  $q = 0.2$ , then  
the expected fraction of false  
discoveries among the rejected is about  
0.2. We rejected  $R = 8$  hypotheses, so  
the expected number of false positives is  
approximately.

$0.2 \times 8 = 1.6 \approx$  about 1-2 false (+ve).

Problem 3 3.2) With impressions multiplied by 10  
( $A=1600, B=1200, C=1500$ ) but clicks  
unchanged (60, 31, 41), all CTRs  
becomes  $\sim 10\times$  smaller, so the  
observed statistic

$$T = \max(\text{CTR}) - \text{mean}(\text{CTR})$$

also becomes about  $10\times$  smaller.

Under the null, the randomness in CTR  
shrinks only like  $1/\sqrt{n}$  when sample  
size  $s$  increase (bigger  $n \rightarrow$  less variability)  
not like  $1/n$ . So the null  
distribution of  $T$  gets tighter, but not  
as fast as the observed  $T$  shrinks in this  
setup.

So, the observed  $T$  is less extreme relative  
to the null spread, meaning the  $p$ -value  
becomes larger. This matches the general  
 $p$ -value idea -  $p$ -value is the  
probability of getting a test statistic  
at least as extreme as observed.